

M Navaneeth Krishna

Senior Software Engineer | ASP.NET Core • Flutter • Data Systems

Indian Institute of Technology Madras — BS Data Science (2025-2029)

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PROFESSIONAL SUMMARY

Full-stack engineer with 4+ years building **production systems across backend and mobile**. Currently architecting a **Zoho Books-style cloud accounting platform** — complete with P&L, Balance Sheet, Trial Balance, double-entry ledger, invoicing, and multi-tenant business management. Designed **Stripe-powered corporate card infrastructure** with real-time transaction processing, spending controls, and automated ledger reconciliation. Experience designing **real-time streaming pipelines**, multi-tenant data architectures, and high-throughput reporting engines.

Creator of open-source tools: **PipeX** (Flutter state management 3.48x faster than Riverpod), **fbloc_event_gen** (70-85% BLoC boilerplate reduction), **ModMon** (ASP.NET project scaffolding CLI), and **MediateX** (MediatR alternative). Deep expertise in fintech, accounting domain logic, payment systems, financial data integrity, distributed systems, and cross-platform mobile development with BLE/IoT integrations.

SKILLS & EXPERTISE

Backend: ASP.NET Core, C#, Python, Entity Framework Core, SQL, REST APIs, SignalR, Stripe API, Background Services

Mobile: Flutter, Dart, BLoC, Riverpod, Provider, GetX, PipeX (custom), BLE, Push Notifications, Local Storage

Databases: MSSQL, PostgreSQL, Redis, Firebase — Data Warehousing, Query Optimization, Indexing, Materialized Views

Architecture: DDD (Bounded Contexts, Aggregates, Domain Events), CQRS, Clean Architecture, Modular Monolith, MVVM, Repository Pattern

Distributed Systems: Multi-Tenant Architecture, Azure Service Bus, Event-Driven Design, SignalR Real-Time Streaming, Message Queues

Accounting Domain: Double-entry ledger, Auto journal posting, Debit/Credit balancing, Period closing, GAAP-compliant reporting, Chart of Accounts

Payment Systems: Stripe Connect, Corporate card issuance, Real-time spending controls, Transaction reconciliation, Webhook processing

Data Processing: Financial pipelines, ETL workflows, Stripe webhook ingestion, Report engines, Batch processing, Redis caching

IoT/Hardware: BLE device communication, Protocol debugging, Connection resilience, Real-time hardware control

DevOps: Docker, Docker Compose, GitHub Actions, Codemagic, Fastlane, Firebase Auth, Containerized Deployments

Performance: Widget optimization, Lazy loading, Memory profiling, Frame rate analysis, Background report generation

OPEN SOURCE LIBRARIES & TOOLS

PipeX — High-Performance Flutter State Management

pub.dev/packages/pipe_x | Benchmark: github.com/navaneethkrishnaindeed/rainbench

- Built pipe-based reactive state management achieving **3.48x faster than Riverpod**, **7.05x faster than flutter_bloc**
- Custom dependency injection with scoped modules, lazy initialization, and lifecycle-aware disposal
- Deployed in 3+ production applications handling complex state flows

fbloc_event_gen — BLoC Code Generator

pub.dev/packages/fbloc_event_gen

- Compile-time code generation eliminating **70-85% of BLoC boilerplate**
- Auto-generates events, states, copyWith, handlers from single annotated class
- Reduced feature implementation from 80-120 lines to 10-20 lines — used across team projects
- Integrated into CI/CD pipelines with `build_runner` for consistent code generation

ModMon — ASP.NET Project Scaffolding CLI

- CLI tool generating production-ready ASP.NET Core projects: `modmon init <project-name>`
- Scaffolds DDD folder structure, Docker Compose, database scripts, EF migrations, and dev utilities
- Pre-configured with authentication, background services, logging, and modular monolith patterns

MediateX — Mediator Pattern Library

- Built **MediatR alternative when MediatR switched to commercial licensing**
- Full mediator pattern for ASP.NET Core — request handlers, notifications, pipeline behaviors

EXPERIENCE

Senior Software Engineer L1

Jan 2025 - Present

GeekyWolf | Trivandrum, Kerala, India

Lead Backend Developer — Cloud Accounting Platform (Zoho Books Alternative)

Financial Reporting & Analytics

- Built real-time **Profit & Loss**, **Balance Sheet**, **Trial Balance**, and **Cash Flow Statement** generation engine
- Implemented **Aged Receivables/Payables** tracking, **Account Reconciliation**, and custom report builder with date-range filters
- Project-wise P&L reports with cost center allocations and billable hours tracking
- Built **financial data pipeline** processing transactions into real-time reports
- Architected **data warehousing layer** for historical financial analytics, aggregated reporting, and trend analysis
- Created **materialized views** and Redis-cached report snapshots for low-latency dashboard queries

Double-Entry Accounting Core

- Complete **Chart of Accounts** with hierarchical account types (Assets, Liabilities, Equity, Income, Expenses)
- **Auto-generated journal entries** for every transaction — invoices, bills, payments, credit notes auto-post to ledger
- Manual journal support with multi-line entries, period closing, and opening balance imports
- Double-entry ledger with automatic journal posting, debit/credit balancing, and period closing

Invoicing, Bills & Documents

- Full invoice lifecycle: create → send → partial payments → credit notes → reconciliation
- Bill management with vendor tracking, due date alerts, and payment matching
- Customizable templates with branding, multi-currency, and tax configurations (GST/VAT)

Stripe Payment Infrastructure

- Architected end-to-end **Stripe Connect** integration for multi-tenant business payment processing
- Built complete **corporate card issuance system** — provision branded cards per business, assign to projects/employees
- Implemented **real-time spending controls**: set/update limits, freeze/unfreeze cards, block categories, enforce policies
- Designed **project-based card allocation** — each project gets dedicated card with isolated budget tracking
- Built **webhook ingestion pipeline** processing card transactions in real-time, auto-posting to double-entry ledger
- Implemented **transaction categorization** and merchant data enrichment for expense analytics
- Created card lifecycle management: issuance → activation → spending → reconciliation → deactivation

Multi-Tenant Enterprise Architecture

- **Business → Users → Roles** permission system with complete data isolation per tenant
- Architected **tenant-isolated data stores** with per-tenant configurations, chart of accounts, and access controls
- Each business: own chart of accounts, tax rules, invoice sequences, and configurations
- Vendor/Customer portals with transaction history, outstanding balances, and statement generation
- Implemented **Role-Based Access Control (RBAC)** with Business → Users → Roles permission hierarchy
- Built **audit trail system** tracking all data mutations with user context, timestamps, and change history

Data Pipeline & ETL Workflows

- Designed **ETL workflows** for bulk data imports (CSV/Excel) — opening balances, contacts, historical transactions
- Implemented **background job processors** for scheduled report generation, email ingestion, and data reconciliation
- **Import/Export**: Bulk CSV/Excel import (opening balances, contacts), PDF/Excel export

Real-Time Streaming & Event Processing

- Implemented **Azure Service Bus** (Kafka-equivalent messaging) for async event processing across distributed services
- Integrated **SignalR** for real-time data streaming — live dashboard updates, notifications, collaborative alerts
- Built **email ingestion pipeline** using Brevo API with background listeners auto-processing incoming invoices
- Implemented **event-driven architecture** with domain events triggering downstream data updates
- **Email Tray**: Brevo + background service auto-captures incoming invoices/bills from email attachments

Flutter Mobile Development

ControlCore — Fintech Accounting App

- Core contributor to Flutter mobile app for cloud accounting platform (invoicing, bills, reports, multi-tenant)
- Integrated PipeX and fbloc_event_gen into codebase, reducing feature development time by 40%
- Implemented offline-first data sync with local caching and background synchronization
- Set up CI/CD pipelines using GitHub Actions for automated builds and Play Store deployment

EverScent — IoT Smart Diffuser App

- Developed complete **Bluetooth Low Energy (BLE)** communication layer for hardware-controlled smart diffusers
- Implemented real-time device pairing, signal strength monitoring, and automatic reconnection handling
- Built scheduling system with fragrance intensity controls, timers, and preset configurations
- Collaborated with embedded firmware team to debug protocol issues and optimize packet transmission
- Added connection state logging and graceful fallback mechanisms for unstable BLE environments

Sr. Flutter Developer / Sr. Software Developer

Feb 2024 - Jan 2025

Ivory Technologies Pvt. Ltd. | Trivandrum, Kerala, India

- Architected scalable Flutter apps using clean architecture with feature-based modularization
- Developed enterprise mobile applications with clean architecture patterns
- Implemented offline-first data sync with local caching, background synchronization, and conflict resolution
- Implemented **local caching strategies** with background sync for intermittent connectivity
- Set up Fastlane + Codemagic for automated iOS/Android builds and TestFlight distribution
- Set up CI/CD pipelines for automated builds, testing, and deployment workflows
- Established code review standards and architecture guidelines for the mobile team
- Integrated REST APIs and implemented offline-first data synchronization

Flutter Developer / Software Developer

Feb 2023 - Jan 2024

Gmagica | Delhi, India

- Built cross-platform mobile applications consuming ASP.NET Core backends
- Built cross-platform mobile apps with complex UI animations and custom widgets
- Integrated REST APIs with proper error handling, caching, and retry mechanisms
- Collaborated with design team to implement pixel-perfect responsive layouts
- Collaborated with backend teams on API design and data contract specifications

Mobile Apps Engineer (Flutter) / Software Engineer

Aug 2022 - Mar 2023

Ykipp (Talrop) | Malappuram, Kerala, India

- Built Flutter apps with clean architecture and conducted code reviews
- Clean architecture implementation, code reviews, team mentoring
- Agile development practices and sprint planning

Junior Project Coordinator

Oct 2021 - Aug 2022

Tegain | Irinjalakuda, Kerala, India

- Project management and stakeholder coordination
- Stakeholder communication and Agile sprints management

EDUCATION

BS in Data Science and Applications — Indian Institute of Technology Madras

2025 - 2029